1. **Can data in memory be called a file? Explain**

A file is data that is stored permanently on a storage device, like a hard drive or SSD, while memory (RAM) only stores data temporarily while the program is running. Once the computer is turned off, the data in memory is lost, but a file stays saved. According to A Guide to Programming in Java Third Edition, memory is used to hold data and instructions while the CPU is working, while files are used for permanent storage.

2. **Write the import statement required to access the file class in an application**

import java.io.File;

The File class is in the java.io package, so this import lets your program create and use file objects such as checking if a file exists, creating a file path, or getting file information.

3. **Identify the error in the following statement: File textFile = new File("c:\inventory.txt");**

There is only one slash in it making the program unable to find out what the file is it also doesn't have the two dots before it
File textFile = new File("..c:\\inventory.txt");

4. **Which statement is used to write an exception handler**
    a) catch (IOException e)

    The catch block is what handles the error if one happens inside the try block.

    A Comp Sci 30 way to explain it:
    The catch statement tells the program what to do when an error happens, instead of letting the program crash.

    b) Write an exception handler to handle an IOException if a specified file name cannot be used to create a file. The exception handler should display appropriate messages to the user.

```
try {
   File textFile = new File(fileName);
   textFile.createNewFile();
}
catch (IOException e) {
   System.out.println("The file could not be created.");
   System.out.println("Please check the file name and try again.");
}
```

5. a) What is the name of the stream for displaying error messages

System.err.println("File could not be opened.");

System.err is a special output stream in Java that is specifically for error messages, separate from the normal output stream System.out. This helps the program clearly show problems or errors to the user without mixing them with regular program messages. For example, if a file cannot be opened or created, messages sent to System.err make it easy to see that an error occurred. It is commonly used in debugging and when handling exceptions.

b) Where are these messages displayed

Messages sent to System.err are displayed in the console or terminal window where the Java program is running. This stream is specifically used for error messages, separate from normal program output sent through System.out. Although both streams usually appear in the same console, System.err allows some environments to highlight or distinguish errors, making it easier for users and programmers to notice problems. For example, if a program tries to open a file that doesn't exist and System.err.println("File not found") is used, the message appears clearly as an error, helping identify and fix issues quickly.

6. a) what does the file stream keep track of?

A file stream in Java keeps track of where you are in the file and what data has been read or written.When a program reads from or writes to a file, the file stream remembers the current position in the file so it knows which part of the file to read or write next. It also keeps track of the type of operation (input or output) and handles buffering, which makes reading and writing more efficient. This allows the program to work with files sequentially without losing track of data.

b) What characters together make up a line terminator

A line terminator tells the program that the current line has ended and the next characters belong to a new line. When reading or writing files, Java recognizes these characters to separate lines correctly, no matter which operating system the file was created on.

7. What two classes are used together to write data to a file

The two classes commonly used together in Java to write data to a file are FileWriter and PrintWriter. FileWriter connects the program to the file and allows characters to be written, while PrintWriter provides easy-to-use methods like print() and println() for writing text. Together, they let a program write data line by line to a file and close it safely when finished.

8. Write a statement to convert account balances that have been read from a text file to a double value and add them to totalBalance

```
totalBalance += Double.parseDouble(accountBalanceString);
```

9. Explain the difference between object serialization and object deserialization

Object serialization and deserialization are opposite processes in Java for saving and restoring objects. Serialization is the process of converting an object into a stream of bytes so it can be saved to a file or sent over a network, preserving its data and state. Deserialization is the process of reading that stream of bytes and reconstructing it back into a usable object in memory. In other words, serialization lets a program save an object, while deserialization lets it restore the object with all its original information intact.

10. What interface must be implemented if objects of a class are to be in a file

The Serializable interface is a marker interface, which means it does not have any methods. By implementing it, you tell Java that objects of this class can be converted into a stream of bytes (serialized) and later restored (deserialized) from a file. Without this interface, trying to write objects to a file will cause a NotSerializableException.